

Updating parameters

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Gradients with a numerical check

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Loss functions

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Composing layers

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Detect overshooting for Updating parameters. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Updating parameters. Proposed explanation: “Subtracting a negative gradient decreases the parameter”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Update a new model independently

Use $x = 1$, target 4, $w = 1$, $b = 1$, squared loss and learning rate 0.1. Compute prediction, residual, gradient [2 times residual times x , 2 times residual], new parameters and new loss.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Apply a negative gradient

Starting with parameter $a = 2$, gradient -3 and learning rate 0.2, calculate the new a .

My reasoning and evidence

6. W2 Detect overshooting

For the lesson start [1, 0], apply rate 0.3 and calculate prediction and loss.

My reasoning and evidence

7. W3 Preserve one snapshot

Explain why recomputing the bias gradient after changing w is not the simultaneous update specified here.

My reasoning and evidence

8. W4 Update a new model independently

Use $x = 1$, target 4, $w = 1$, $b = 1$, squared loss and learning rate 0.1. Compute prediction, residual, gradient [2 times residual times x , 2 times residual], new parameters and new loss.

My reasoning and evidence

9. W5 Evaluate a zero gradient

At a point with gradient $[0, 0]$, what does this update do? Does that alone prove a global minimum on any loss surface?

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled updating parameters evidence sheet to Image-pattern Network Investigation. Show how you can apply a learning rate to an ordered gradient. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Updating parameters

Use one explicit gradient update and then recompute the loss.

Model $y=2w+b$, target 5. Start $w=1$, $b=1$; learning rate 0.05. Initial $y=3$, error $y-\text{target}=-2$, squared loss 4.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/11

Loss before and after rate 0.05 update

squared error

before



after



Bar length shows the supplied value; the marked baseline is zero.

Updating parameters

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gradient

Gradient for $[w, b] = [2 \times (-2) \times 2, 2 \times (-2)] = [-8, -4]$.



2. Update simultaneously

New $w = 1 - 0.05 \times (-8) = 1.4$; new $b = 1.2$.



3. Check new loss

New $y = 4$, residual -1 , loss 1 : lower than 4 .

CHANGE ONE THING

Set learning rate 0.2 from the same starting point.

Explain the changed result using the matching lesson.

Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.